

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1293PC303 - NETWORK THEORY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

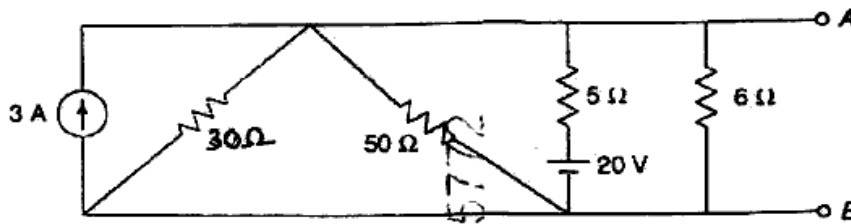
12

1. Which of the following is not an example of a linear element?
a. Resistor b. Thermistor c. Inductor d. Capacitor
If the resistors of star connected system are R_1, R_2, R_3 then the resistance between 3 and 1 in delta connected system will be?
a. $(R_1R_2 + R_2R_3 + R_3R_1)/R_1$
b. $(R_1R_2 + R_2R_3 + R_3R_1)/R_3$
c. $(R_1R_2 + R_2R_3 + R_3R_1)/R_2$
d. $(R_1R_2 + R_2R_3 + R_3R_1)/(R_3+R_1)$
2. Kirchhoff's Current law is based on the law of conservation of _____.
a. energy b. momentum
c. mass d. charge
3. In Superposition theorem, while considering a source, all other current sources are?
a. short circuited b. changes its position
c. open circuited d. removed from the circuit
4. In Reciprocity Theorem, which of the following ratios is considered?
a. Voltage to current b. Current to current
c. Voltage to voltage d. No ratio is considered
5. The maximum power is delivered from a source to its load when the load resistance is _____ the source resistance.
a. greater than b. less than c. equal to d. less than or equal to
6. The current in the R-L circuit at a time $t = 0^+$ is?
a. V/R b. R/V c. V d. R
7. After how many time constants, the transient part reaches more than 99 percent of its final value?
a. 2 b. 3 c. 4 d. 5
8. The time constant of an R-C circuit is?
a. RC b. R/C c. R d. C
9. The s-domain equivalent of the capacitor reduces to a capacitor with impedance?
a. sC b. C c. $1/C$ d. $1/sC$
10. In determining short circuit impedance parameters, among V_1, V_2, I_1, I_2 , which of the following are independent variables?
a. I_1 and V_2 b. V_1 and I_2 c. I_1 and I_2 d. V_1 and V_2

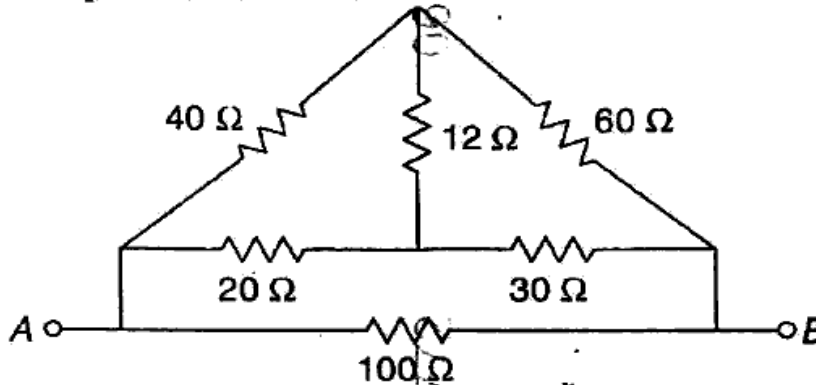
12. The value of one decibel is equal to?
 a. 0.115 N b. 0.125 N c. 0.135 N d. 0.145 N

Q2. Solve the following.

- A) Using source transformation, convert the circuit shown in fig. into single current source and single resistance. 6

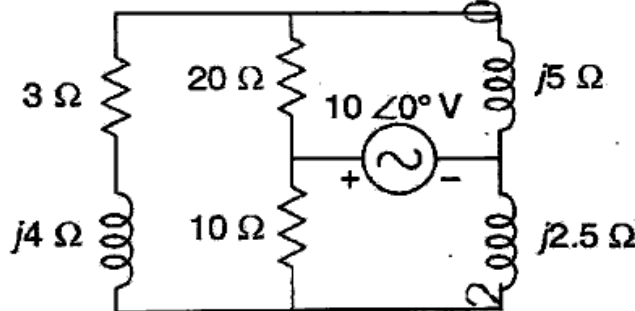


- B) Find the equivalent resistance between A and B in the network of Fig 6

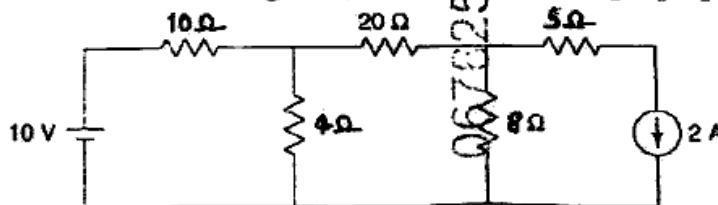


Q3. Solve the following.

- A) Find the current through the $(3 + j4)$ impedance in Fig. using mesh analysis. 6

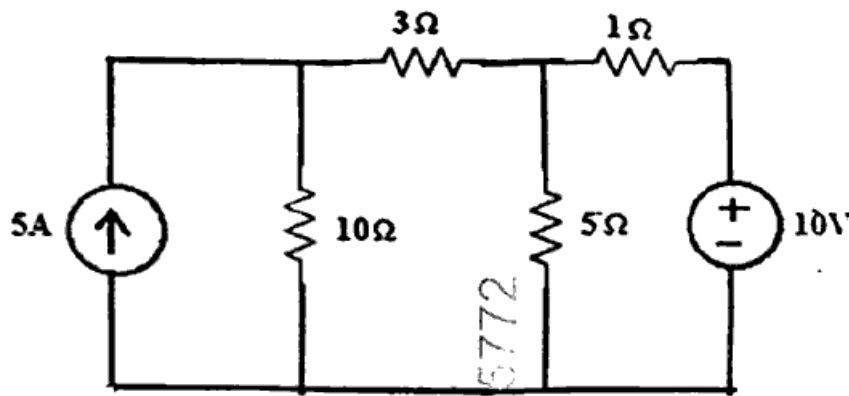


- B) Find the current through the 20 ohm resistor using superposition theorem 6



Q4. Solve Any Two of the following.

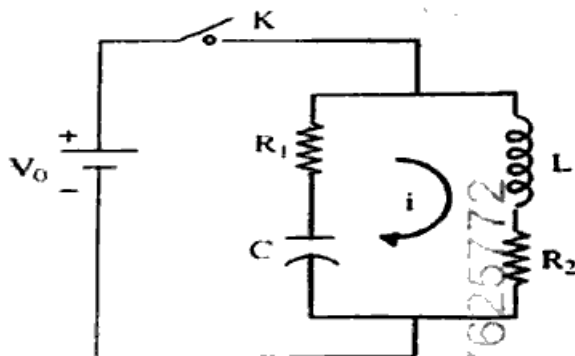
- A) Determine the current in 10Ω resistor for the following network by using nodal analysis 6



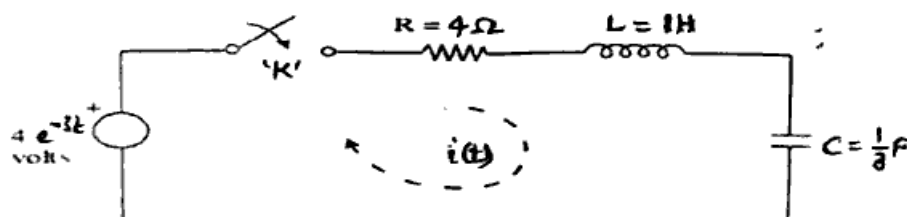
- B) State and prove Tellegen's theorem. 6
 C) A balanced delta-connected load of $(8 + j6)$ ohms per phase is connected to a three-phase, 230 V supply with phase sequence R-Y-B. Find the line current, power factor, power, reactive volt-amperes and the total volt-amperes. Draw the phasor diagram. 6

Q5. Solve Any Two of the following.

- A) Derive the expression for Z-parameters of a two port network and reciprocal condition. 6
 B) The switch K in fig 7(a) is closed at $t = 0$ connecting battery to an unenergized network. Determine : $i(0^+)$, di/dt at $t = (0^+)$. 6

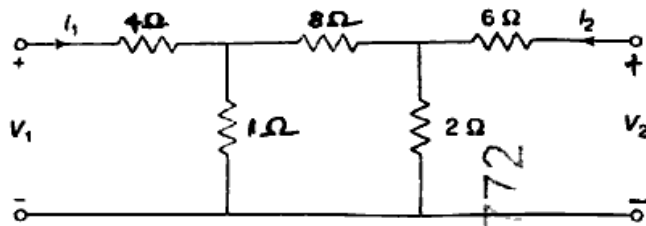


- C) An exponential voltage of $4e^{-3t}$ volts is applied to a series RLC circuit at $t=0$. Obtain particular solution for current $i(t)$ use Laplace Transform method At $t = 0$, switch 'K' is closed refer 6



Q6. Solve Any Two of the following.

- A) Derive the Expression for the impulse response of series RC Network using of Laplace transform. 6
- B) Determine ABCD-parameters for Network shown in Fig. 6



- C) Explain series resonance and determine resonant frequency, half power frequency, quality factor and bandwidth. 6

*** End ***